



Getting started

This guide will instruct you through:

- Creating your first [R2 bucket](#) and enabling its [data catalog](#).
- Creating an [API token](#) needed for query engines to authenticate with your data catalog.
- Using [Pylceberg ↗](#) to create your first Iceberg table in a [marimo ↗](#) Python notebook.
- Using [Pylceberg ↗](#) to load sample data into your table and query it.

Prerequisites

1. Sign up for a [Cloudflare account ↗](#).

2. Install [Node.js](#).

› Node.js version manager

1. Create an R2 bucket

Wrangler CLI

Dashboard

1 If not already logged in, run:

```
npx wrangler login
```

2 Create an R2 bucket:

```
npx wrangler r2 bucket create r2-  
data-catalog-tutorial
```

2. Enable the data catalog for your bucket

Then, enable the catalog on your chosen R2 bucket:

```
npx wrangler r2 bucket catalog enable r2-  
data-catalog-tutorial
```

When you run this command, take note of the "Warehouse" and "Catalog URI". You will need these later.

3. Create an API token

Iceberg clients (including [Pylceberg ↗](#)) must authenticate to the catalog with an [R2 API token](#) that has both R2 and catalog permissions.

- 1 In the Cloudflare dashboard, go to the **R2 object storage** page.

Go to **Overview** [↗](#)

- 2 Select **Manage API tokens**.
- 3 Select **Create API token**.
- 4 Select the **R2 Token** text to edit your API token name.
- 5 Under **Permissions**, choose the **Admin Read & Write** permission.
- 6 Select **Create API Token**.
- 7 Note the **Token value**.

4. Install uv

You need to install a Python package manager. In this guide, use [uv ↗](#). If you do not already have uv installed, follow the [installing uv guide ↗](#).

5. Install marimo and set up your project with uv

We will use [marimo](#) as a Python notebook.

- 1 Create a directory where our notebook will be stored:

```
mkdir r2-data-catalog-notebook
```

- 2 Change into our new directory:

```
cd r2-data-catalog-notebook
```

- 3 Initialize a new uv project (this creates a `.venv` and a `pyproject.toml`):

```
uv init
```

- 4 Add marimo and required dependencies:

Python

```
uv add marimo pyiceberg pyarrow  
pandas
```

6. Create a Python notebook to interact with the data warehouse

- 1 Create a file called `r2-data-catalog-tutorial.py`.
- 2 Paste the following code snippet into your `r2-data-catalog-tutorial.py` file:

Python

```
import marimo
```

```
__generated_with = "0.11.31"
```

```
app = marimo.App(width="medium")
```

```
@app.cell
```

```
def _():
```

```
    import marimo as mo
```

```
    return (mo,)
```

```
@app.cell
```

```
def _():
```

```
    import pandas
```

```
    import pyarrow as pa
```

```
    import pyarrow.compute as pc
```

```
    import pyarrow.parquet as pq
```

```
    from pyiceberg.catalog.rest import  
RestCatalog
```

```
    # Define catalog connection details  
(replace variables)
```



```
WAREHOUSE = "<WAREHOUSE>"
TOKEN = "<TOKEN>"
CATALOG_URI = "<CATALOG_URI>"

# Connect to R2 Data Catalog
catalog = RestCatalog(
    name="my_catalog",
    warehouse=WAREHOUSE,
    uri=CATALOG_URI,
    token=TOKEN,
)
return (
    CATALOG_URI,
    RestCatalog,
    TOKEN,
    WAREHOUSE,
    catalog,
    pa,
    pandas,
    pc,
    pq,
)
```

@app.cell

```
def _(catalog):  
    # Create default namespace if needed  
  
    catalog.create_namespace_if_not_exists("  
    default")  
  
    return
```

```
@app.cell  
def _(pa):  
    # Create simple PyArrow table  
    df = pa.table({  
        "id": [1, 2, 3],  
        "name": ["Alice", "Bob",  
        "Charlie"],  
        "score": [80.0, 92.5, 88.0],  
    })  
    return (df,)
```

```
@app.cell  
def _(catalog, df):  
    # Create or Load Iceberg table  
    test_table = ("default", "people")
```

```
        if not
catalog.table_exists(test_table):
            print(f"Creating table:
{test_table}")
            table = catalog.create_table(
                test_table,
                schema=df.schema,
            )
        else:
            table =
catalog.load_table(test_table)
        return table, test_table
```

```
@app.cell
def _(df, table):
    # Append data
    table.append(df)
    return
```

```
@app.cell
def _(table):
    print("Table contents:")
    scanned = table.scan().to_arrow()
```

```
print(scanned.to_pandas())  
return (scanned,)  
  
@app.cell  
def _():  
    # Optional cleanup. To run uncomment  
    and run cell  
    # print(f"Deleting table:  
    {test_table}")  
    # catalog.drop_table(test_table)  
    # print("Table dropped.")  
    return  
  
if __name__ == "__main__":  
    app.run()
```

3 Replace the CATALOG_URI, WAREHOUSE, and TOKEN variables with your values from sections **2** and **3** respectively.

4 Launch the notebook editor in your browser:

```
uv run marimo edit r2-data-catalog-tutorial.py
```

Once your notebook connects to the catalog, you'll see the catalog along with its namespaces and tables appear in marimo's Datasources panel.

In the Python notebook above, you:

1. Connect to your catalog.
2. Create the default namespace.
3. Create a simple PyArrow table.
4. Create (or load) the people table in the default namespace.
5. Append sample data to the table.
6. Print the contents of the table.
7. (Optional) Drop the people table we created for this tutorial.

Learn more

Managing catalogs →

Enable or disable R2 Data Catalog on your bucket, retrieve configuration details, and authenticate your Iceberg engine.

Connect to Iceberg engines →

Find detailed setup instructions for Apache Spark and other common query engines.

Was this helpful?



Previous

← R2 Data Catalog

Next

Manage catalogs →

Last updated: Sep 25, 2025

Resources

[API](#)

[New to Cloudflare?](#)

[Directory](#)

[Sponsorships](#)

[Open Source](#)

Company

[cloudflare.com](#)

[Our team](#)

[Careers](#)

Community

 [X](#)

 [Discord](#)

 [YouTube](#)

 [GitHub](#)

Support

[Help Center](#)

[System Status](#)

[Compliance](#)

[GDPR](#)

Tools

[Cloudflare Radar](#)

[Speed Test](#)

[Is BGP Safe Yet?](#)

[RPKI Toolkit](#)

[Certificate Transparency](#)

© 2025 Cloudflare, Inc. • [Privacy Policy](#) • [Terms of Use](#)

- [Report Security Issues](#)
- [Trademark](#)
-  [Cookie Preferences](#)